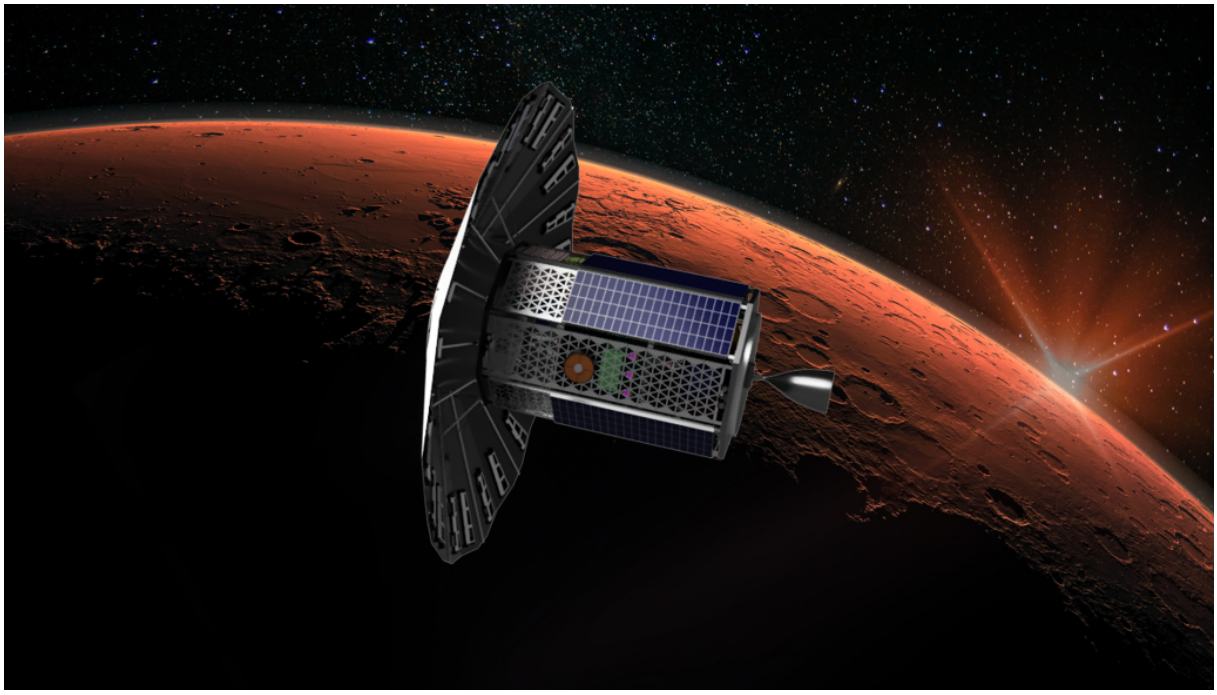
Aerocapture Mission to Mars

Exploration of a fuel-efficient, accurate insertion with a focus on the martian sphere of influence

Mission Design & Analysis ADM02



Academic responsible: Dr Paul Bruce
Supervisor: Dr Chris Cantwell

Department: Department of Aeronautics
Course: H401/H415 MEng Aeronautical Engineering
Module: AERO60004 - Group Design Project
Academic year: 2021/2022

Student: Yago Molano Gomez
CID: 01714721
Date: 20/06/2022

Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ
U.K.

Sub-Group	Name	Role
ADM01: Project Coordination	Camille Boutrolle Jia Hui Sofia Tan Jinlun Jimmy Zhang	Project Coordination
ADM02: Mission Design & Analysis	Aaditi Vaze Luca Mazzotta Mohit Tank Yago Molano Gomez	Interplanetary Trajectory Design & Optimisation Interplanetary Trajectory Design & Optimisation Mission Architecture & Future Roadmap Hyperbolic Trajectory Design & Modelling
ADM03: Spacecraft Systems & Controls	Chorley King Dannielle Lee Lucas Parkins Nihal Simha	Command & Data Handling & Power Systems Guidance & Navigation Telemetry Tracking & Command Attitude Determination & Control
ADM04: Spacecraft Static Structural Design	Samuel Cheung Sze Hang Prudence Lee	Static Structural Design & CAD
ADM05: Spacecraft Dynamic Structural Design	Julia Wiszniowska Luc Groshens Luke Wyatt	Deployable Panel System Structural Design Deployable Rib & Spar System Structural Design Deployable Mechanism Structural Design & TPS
ADM06: Spacecraft Aerothermal Design	Claire Namuroy Erikas Kymantas Jaime Galiana Herrera Segundo Urraza Atue	Heat Transfer & Pressure Modelling & CFD Atmospheric & Space Flight Dynamics Hypersonic Stability Analysis Atmospheric & Space Flight Dynamics
ADM07: Spacecraft & Environment Interaction	Jae Min Oh Livia Trambaiolo Mohammed Choudhury Yasmine Chan	Radiation & Debris Protection Instrumentation & Scientific Payload Thermal Management Emergency Planning

Abstract

The mission characterization process quantified the requirements, identified possible system architectures, defined the system drivers and selected a final architecture. The use of an aerocapture maneuver requires high accuracy in flight path angle whilst minimising fuel-intensive maneuvers. To achieve this, an unconventional backwards adaptation of the method of patched conics is explored in this report. The report specially focuses on the challenges imposed by this approach on the hyperbolic orbit towards mars and its integration with the interplanetary and aerothermal models to optimise from launch to final orbit. It calculates the decrease in delta-V from the use of aerocapture and how it can be further reduced in the future. The report finishes with the method's shortcomings and future steps.

Acknowledgements

I would like to thank the following persons for their help with my project:

- **Dr Chris Cantwell** for his support and advice throughout the project, as well as for his helpful, continuous feedback that helped me improve week after week.
- **Colleagues** for all of their hard work, contagious enthusiasm and eagerness of helping you tackle a hard problem.
- **Dr Davide Amato** for his clarification on orbital mechanics concepts, and willingness to meet and help even though he was not our supervisor.
- **The other professors involved** for their useful feedback during the presentations that guided me towards a better final design.

Contents

List of Figures iii

List of Tables iv

1 Introduction 1

1.1 Mission Characterization . . . 1

1.1.1 Preliminary Needs, Requirements and Constraints . . . 1

1.1.2 Alternative mission concepts and architectures . . . 2

1.1.3 Key System drivers . . . 2

1.1.4 Final architecture . . . 3

1.2 Motivation . . . 3

2 Method of patched conics 4

2.1 Theoretical Background . . . 4

2.2 Approach taken . . . 5

3 Hyperbolic orbit in perifocal frame 6

3.1 Initial set of simultaneous equations . . . 6

3.2 Backwards approach: Calculating FPA from r_p and v_{inf} . . . 6

3.3 Using a Non-Linear Least Squares Solver to invert the function . . . 7

3.4 Propagation of the orbit . . . 7

4 Integration and Interfacing 8

4.1 Patching to Interplanetary Trajectory . . . 8

4.2 Input to Aerothermal team's model . . . 8

4.3 Post Aerocapture: Transforming output to orbital elements . . . 10

5 Discussion 11

5.1 Uncertainty propagation throughout the trajectory and its potential effect on mass estimates . . . 11

5.2 Inter-dependency of variables and consequent mismatch between models . . . 11

List of Figures

1 Figure showing how the interplanetary and hyperbola are patched in the limit of the sphere of influence. [1] 4
2 Figure showing key geometrical parameters of hyperbolic orbit [2] 5
3 Figure showing the relationship between Δ and r_p [1] 5
4 Figure showing intersection point between hyperbolic function and circular function. . 7
5 Graphical representation of Centered Mars Mean Equator and IAU-Vector of Date frame. [3] 9
6 Hyperbolic insertion showing perturbed and unperturbed trajectory 9
7 Diagram showing initial approach (straight lines) compared to current identified interdependencies;. 12

List of Tables

1 Table showing main requirements and constraints for the two missions. 1
2 Key system drivers and their relative importance 2
3 Key elements of the chosen architecture for the standalone mission 3
4 Orbital elements determined from FPA 7
5 Remainder of orbital elements and time of arrival 8
6 Insertion coordinates for aerothermal model. 10
7 Orbital elements after aerocapture 10
8 Table showing the total delta-V needed for our current iteration, a potential better optimization, and no aerocapture 11

1 Introduction

The introduction includes a summary of the mission characterization process, as well as the motivation and objectives for the trajectory design.

1.1 Mission Characterization

During the first two weeks, the work of the Mission Design Team focused on:

- Defining preliminary mission needs, requirements and constraints.
- Defining alternative mission concepts and architectures.
- Defining key system drivers.
- Shortlisting and selecting the system architecture.

In this report, needs and mission drivers will be looked at in-depth and a summary of the other points will be included, for which an in-depth analysis can be found in reports [4], [5] and [6].

1.1.1 Preliminary Needs, Requirements and Constraints

After the broad objectives were defined by the ESA Brief [7] and the Project Coordination team, Mission Analysis translated these objectives into preliminary sets of numerical requirements, which may be found in table 1. Many requirements were defined by the brief, but some key requirements were defined by the team.

Table 1: Table showing main requirements and constraints for the two missions.

Scenario	Requirement	Value
1 - Standalone	Entry Mass	<400kg
	Diameter	<2.6m
	Payload Mass	>=20kg
2 - Piggybag	Entry Mass	75-100kg
	Diameter	<1.6m
	Payload Mass	6kg
Generic	Launch Window	2031 - 2033
	Launchers	Ariane 6(2,4), Vega C
	Orbit	e <0.05; alt_p = 500-1000km ; i >70deg;
	Hyperbolic excess speed	3-4 km/s
	Mission design life	2+ years
	Risk	70
	Survivability	Survive the natural environments of mission
	Flight path angle dispersion	<= 0.4 deg
	Payload temperature	TPS keeps payload between -30 and 51 ^{o}C
	Structure integrity	Must withstand loads imparted by Ariane 62.

An example was risk. This is an important requirement to quantify, as it can help guide decisions when choosing one design over another. It is especially important for subsystem-specific trade-off studies at later stages, when this risk can be quantified. Due to the nature of the mission, a technology demonstrator, there was a higher appetite for risk than with other missions. This was due to there being a smaller cost to the risk; for ESA there is no returns even if the mission is successful, except for potential savings in future missions if this technology is proven feasible. Higher risk allows working on more innovative solutions, increasing the potential perceived benefits. A value of 70% was set, as it will allow the mission to work on innovative solutions without a restrictive success percentage, whilst ensuring that the likeliness of success, with the potential perceived benefits for future missions, is

enough to make ESA fund the mission. This is lower than other unmanned missions using conventional technologies, which usually have 90% or higher success rates. [8]. However, this number will solely be used qualitatively at this stage, and should be refined in the future.

A few other subsystem-specific requirements were set. The thermal protective system must maintain payload temperature between -30 and 51 $°C$. The environment team must be able to survive the natural environments of the mission, most critically the time spent in LEO and the solar flares expected during transit. Finally, a mission design life was set. As the brief requires to measure and relay data for a year after the final orbit, the mission design life must be at least the transfer time between Earth and Mars plus one year. As the transfer time is expected to be around 300 days, and the period of the final orbit will be less than 2.4526 hours (period of an elliptical orbit with a semi-major axis equal to altitude of 1000km, which is the maximum value for semi-major axis for the final orbit), it was set as a minimum mission life of two years.

1.1.2 Alternative mission concepts and architectures

There were two main mission concepts that were preconceived from the ESA white paper. For the standalone mission, the concept revolved around having a usual mission to mars but substituting decelerating burn for areocapture followed by a short burn to circularise it. The piggyback mission followed a similar concept, with the exception that there were more constraints in the direction of the velocity magnitude imparted, and gravity assisted maneuvers would most likely need to be used. An alternative mission concept was suggested by the team, which was using aerocapture to slow down significantly but not fully, and then use a number of aerobraking maneuvers to lose the velocity completely.

A list of possible architectures identified by each sub-team may be found on figure [4].

1.1.3 Key System drivers

After the possible array of architectures were brainstormed, the Mission Analysis team identified the principal mission parameters that would measure the success of the mission, and how they affected other indicators. A list of key system drivers may be found on table 2.

The most obvious success parameter for this mission is the percentage mass of fuel. Aerocapture technology has the potential to slash the conventional value of around 70% propellant of mass [8] to a fraction of this value. Therefore, the lower the value the most successful the mission can be considered. However, this neglects the effect the different types of aerocapture can have on other components, such as structural weight; potentially the loads might vary as well the heat to be dissipated by the TPS, varying the weight. Therefore, a better performance indicator is the percentage weight of payload, as maximising this value will mean minimizing the cost of launch to Mars. This is the main and most important system driver.

Other usual main drivers for space missions are technology readiness, risk, or cost. For a proof-of-concept mission, technology readiness can be considered a bad measure, as most architectures will have low technological readiness and a secondary objective of the mission is to mature these novel concepts. With regards to risk, this is a good systems driver as the mission team will want to choose architectures with a lower risk, to maximise the possibilities of mission success. However, at this early stage, it is extremely difficult to measure risk quantitatively and therefore does not provide a good parameter for comparison.

Table 2: Key system drivers and their relative importance

System drivers	Importance (%)
Payload mass fraction	50
Risk and reliability	20
Cost	20
Interplanetary fuel weight	10
Future potential	N/A

Another parameter to take into account is delta-v for launch. If this delta-v increases significantly when compared to similar missions, the payload mass fraction at launch could be lower than a conventional mission even if this value is higher at mars, and therefore result in an overall more costly mission than a conventional method. Therefore this is an important parameter to look at when choosing the architecture.

Finally, whilst lowering the cost of the mission could be a good secondary objective, as it means you can get more payload to mars and at a cheaper price, it is not as important as the mass fraction payload you can get to Mars. This is because cost is expected to be higher at a proof-of-concept stage and will decrease over time. Nevertheless, cost gives a good combination between fuel savings, as that would reduce the cost of the mission, delta-v of launch, and complexity of the design. Because of this, it was also included.

Two other mission drivers are related to the reliability of the aerocapture trajectory, and the potential for future missions. The accuracy of aerocapture trajectory is an essential driver. If the system is more inaccurate, even if the previous identified system drivers suggest the payload mass will be lower, the uncertainty introduced by a more unreliable method means that extra fuel will need to be carried, lowering the advantages with respect to other options. Some of the options identified for aerocapture, like the ones that are continously varying, have higher potential for future missions as they provide a higher level of control, and could even be used in the future for an unknown atmosphere. Whilst it cannot be calculated quantitatively, it is a sound qualitiative argument that was kept in mind when shortlisting architectures.

It is important to note that most of the system drivers are directly related to the aerocapture method, as this is the most critical subsystem, and once it has been chosen, the best choice for other subsystems is clear.

1.1.4 Final architecture

The final architectures may be found in figure 3. The justification for this selection and shortlisting of mission concepts may be found in [4].

Table 3: Key elements of the chosen architecture for the standalone mission

Geometry	Deployable umbrella
Control system	Hot gas thrusters + gyroscopes
TPS	Woven Carbon Fibre
Space prop	Mono-prop hydrazine
Risk	High
Technology Readiness Level	Low

1.2 Motivation

The use of an Aerocapture maneuver has the potential to greatly increase the percentage of payload mass. This will decrease the cost of getting payload to mars, increase the weight of the objects that can be taken, and as a result bring closer the objective of having a human presence in the red planet.

Nevertheless, an aerocapture maneuver has many challenges, one of which is that the flight path angle (FPA) of entry is a very sensitive parameter. A difference of less than one degree in the velocity angle can cause the spacecraft to crash into mars, or to bounce and fly away in an unbounded. Because of the high velocities involved, changing the velocity vector is extremely costly in terms of propellant, and therefore targeting the neccessary FPA accurately becomes imperative.

This shows how critical it is for Mission Analysis to do a good job with the design of the orbital trajectory. It is essential for Mission Analysis to find a way of targetting the required FPA, maximizing the accuracy of insertion whilst minimising fuel-intensive maneuvers.

2 Method of patched conics

The method of Patched conics was chosen to design the trajectory. This section will give a brief summary of the theory behind the patched conic approach with a focus on the hyperbolic trajectory around mars, and then outlining our particular approach given the accuracy in FPA required.

2.1 Theoretical Background

In essence, the method of patched conics separates the orbit into three parts. The first one, in which the spacecraft is considered to be inside Earth's sphere of influence (SOI), a second interplanetary section in the sun's SOI, and the final one in Mars' SOI. Within the SOIs, it is assumed that only that given planet is influencing the orbit for the spacecraft, and therefore it follows an unperturbed Keplerian path around the planet. [1]. The SOI's radius r_{SOI} for the planets is given by equation 1. Due to the velocities involved, the first and last conics, Earth's and Mars', have a hyperbolic trajectory, whilst the interplanetary trajectory is elliptical.

$$\frac{r_{SOI}}{R_s} = (\frac{m_p}{m_s})^{2/5} \approx 578000km \tag{1}$$

For the interplanetary trajectory, the solutions to Lambert's problem [9] at different transfer times t, starting and final positions are found, which generates pork chops. This can then be used to find which is the best orbit given our requirements. More detail on this method may be found on reports [6] and [5]. Then, the departure and arrival velocities can be computed, to give the starting/final conditions for the hyperbolic conics. Figure 1 shows how the two conics are patched.

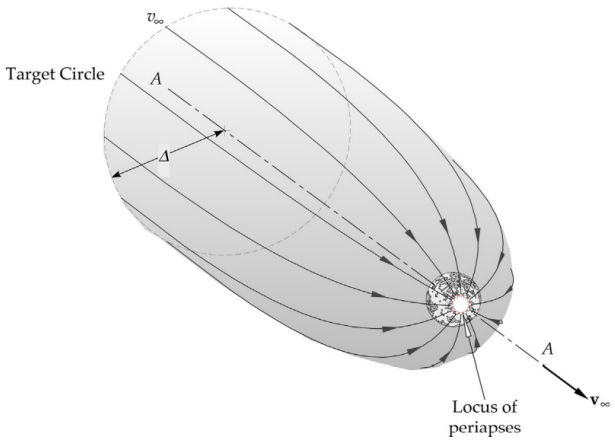


Figure 1: Figure showing how the interplanetary and hyperbola are patched in the limit of the sphere of influence. [1]

The geometry of the hyperbolic trajectories can be observed in figure 2. The energy of the orbit is given by the excess velocity v_∞, which is the velocity relative to Mars outside its SOI. As energy is constant, this allows us to find the velocity as a function of radius, as kinetic energy will equal potential energy. As energy also defines the semi-major axis a, it defines part of the geometry. This is shown in equation 2.

$$\frac{v_\infty^2}{2} = \frac{\mu}{2a} = \frac{v^2}{2} - \frac{\mu}{r} \tag{2}$$

For a hyperbolic trajectory, radius is given by equation 3, which as shown means the periapsis r_p is a function of a and eccentricity e. The final geometric values b and Δ, the offset by which the asymptote misses the planet, are given by equation 4. Note that b and Δ are equivalent, which can be deduced geometrically. These equations fully define the geometry of the hyperbolic trajectory and relate it to v_{inf} and Δ, as shown by figure 3.

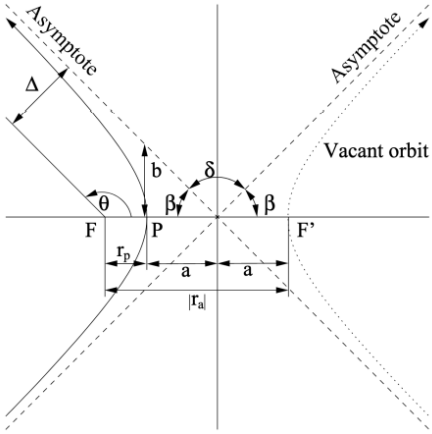


Figure 2: Figure showing key geometrical parameters of hyperbolic orbit [2]

$$r_p = \frac{h^2}{\mu}\frac{1}{1+ecos(\theta_p)} = \frac{a(1-e^2)}{1+e} = a(e-1) \tag{3}$$

$$b = \Delta = atan(\beta) = a\frac{sin(180-\theta_\infty)}{cos(180-\theta_\infty)} = a\frac{\sqrt{\frac{e^2-1}{e}}}{\frac{1}{e}} = a\sqrt{e^2-1} \tag{4}$$

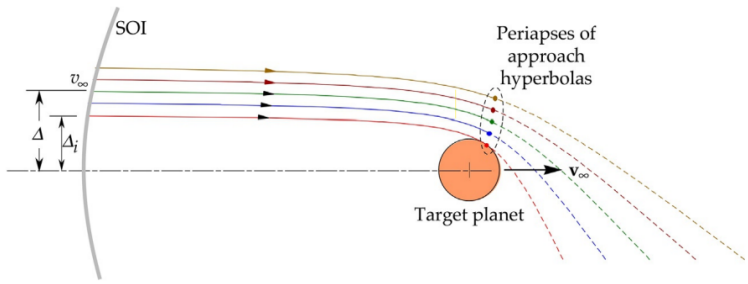


Figure 3: Figure showing the relationship between Δ and r_p [1]

2.2 Approach taken

Traditionally, in the method of patched conics the first step is to compute the heliocentric, interplanetary trajectory from Earth to Mars. As previously described, solving Lambert's problem for a range of dates can be used to generate pork chop plots. Once these have been plotted, a transfer which minimises the delta-V is selected. Once this has been finalised, the needed v_∞ from Earth is used to design the magnitude and direction of the burn. Similarly in Mars, the magnitude of the velocity outside it's SOI is used as the velocity magnitude at asymptote, and Δ at that point is used to define the periapsis radius which allows to calculate all the other orbital elements in the perifocal frame. This is shown visually in figure 1. It is also interesting to note how depending on the point of the circle the planet will be reached from, a different inclination can be achieved, which will allow the team to achieve the required inclination set by section 1.1.1.

As a high level of accuracy is needed in the flight path angle insertion, a non-conventional, backwards approach was taken for our design. The main idea was to use the FPA required by the aerothermal team as the unique input to our code, and then optimise the trajectory to minimize the fuel for transit. To do this, the hyperbolic orbit was to be defined using the FPA, and give the interplanetary team Δ and v_{inf}, or another equivalent set of values, to use as their final conditions. The interplanetary trajectory would then be computed to achieve these conditions. This approach is significantly more challenging than the usual approach, as it involves many iterations in order to get the right results "backwards".

In addition to this, the interplanetary trajectory would be optimised to minimize delta-V near Earth as well as deep space maneuvers, as they are the delta-V's which will utilize fuel. Minimizing them will reduce to the minimum the mass percentage of fuel and cost, and was therefore considered the best approach to maximise the key system drivers. In reality, FPA is not enough to define the hyperbolic orbit, so v_{inf} was also required as input. This was seen as an advantage as it gave more flexibility to the interplanetary trajectory design, which was then optimised to minimise the same delta-V's, but ensuring it didn't lead to a hyperbolic excess velocity outside the constraints set by 1.1.1.

3 Hyperbolic orbit in perifocal frame

3.1 Initial set of simultaneous equations

The flight path angle is defined as the angle between the velocity vector and the plane tangential to the Martian surface at the given point. In the perifocal frame, this is equivalent to the angle between the edge of the Martian atmosphere and the hyperbolic trajectory at the point of intersection, and therefore it is a function of the gradient of the circle and the gradient of the hyperbolic trajectory. The gradient of the circle can be considered to be a function of the eccentric anomaly (E), or of the intersection coordinates (x,y) . The gradient of the hyperbolic trajectory will be a function of θ or (x,y), as well as other elements yet to be defined such as r_p. This will lead to a system of 6 simultaneous equations, without a unique solution. After solving numerically, there was poor convergence in the model, and many results were imaginary or physically impossible. Because of this, another, more simple method was desired.

3.2 Backwards approach: Calculating FPA from r_p and v_{inf}

As stated in section 3, the system of simultaneous equations was not considered a robust and reliable method that could define a singular hyperbolic orbit. To solve this, the approach taken was to create a function that could find the FPA given r_p and v_{inf}, to then invert it.

In order to do this, the first step was to define the hyperbolic trajectory geometrically. Firstly, equation 2 was used to find the semi-major axis from v_{inf}. After this was done, e could be found by rearranging equation 3 as it is only a function of r_p and a . Finally, b was found from equation 4 which fully defines the geometry of the hyperbolic orbit.

The next step was to find the intersection between the circle outlining the atmosphere and the hyperbolic orbit. To do this, symbolic functions were written for both the hyperbolic orbit, shown in equation 5 and the circle, shown in equation 6. They were defined so that the vertical centre of symmetry of the hyperbolic orbit crossed through the origin, as shown in figure 2 and following convention.

$$(\frac{x}{a})^2 - (\frac{y}{b})^2 = 1 \tag{5}$$

$$(x + a + r_p)^2 + y^2 = r_{atm}^2 \tag{6}$$

Once defined, the intersection points were found by equating the symbolic functions. It is important to understand that the hyperbolic function also includes a line in the right half plane, labeled vacant orbit in figure 2, which potentially could lead to up to four intersection points. However this system would be physically impossible, as it requires that the circle has positive x values, which would require that the radius of the atmosphere is larger than the radius of the periapsis plus semi-major axis. From the two intersection values, which due to symmetry are equivalent except for the sign of the y value, the negative was taken, defining that the spacecraft would arrive from the downwards direction. This is a trivial decision, as the geometry that is defined can be rotated fully to exactly match the entry conditions.

Once the intersection point was known, the gradient of both analytical functions was calculated, and the angle between it computed, giving FPA as a function of r_p, as well as defining all orbital elements in the perifocal frame.

3.3 Using a Non-Linear Least Squares Solver to invert the function

Once the function described above was written, it was inverted by creating a function handle which used r_p as input and subtracted the required flight path angle. Then, *lsqnonlin* function was used to find the value for which it is zero. The surface of mars and edge of the atmosphere were set as the bounds for the calculated r_p. This method gave an accuracy of more than 5 decimal places and only took a few fractions of a second to run, so it was chosen as it was the best method of the ones considered to invert this function. Figure 4 show the intersection between the atmosphere and trajectory of the geometrical functions. Table 4 shows the orbital elements outputted from the function, which fully define the hyperbolic orbit in the perifocal frame.

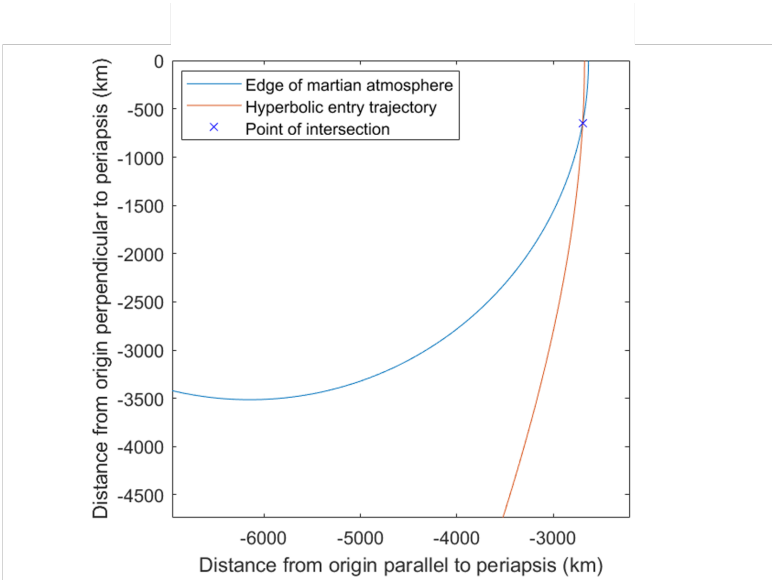


Figure 4: Figure showing intersection point between hyperbolic function and circular function.

Table 4: Orbital elements determined from FPA

Orbital element	Value
a (km)	3.5569E+03
e (deg)	1.9751
θ (deg)	-10.9826

3.4 Propagation of the orbit

Initially, the orbit was propagated in the unperturbed state by applying equations 7 and 8. The time, relative to the point of insertion was calculated using Barker's equation (9), where τ_{int} is found using the same equation with the true anomaly at intersection [10].

$$r = \frac{h^2}{\mu}\frac{1}{1+ecos(\theta)}[cos(\theta)\boldsymbol{i}_e + sin(\theta)\boldsymbol{i}_p] \tag{7}$$

$$v = \frac{\mu}{h}[-sin(\theta)\boldsymbol{i}_e + (e+cos(\theta))\boldsymbol{i}_p] \tag{8}$$

$$\frac{\mu^2(t-\tau_{int})}{h^3} = \frac{1}{2}tan\frac{\theta}{2} + \frac{1}{6}tan^3\frac{\theta}{2} \tag{9}$$

After plotting the results and checking it matches the symbolic function, a perturbed model was studied. Cowell's method was applied, which involves putting Euler's equation (10) in state-space form. The only perturbations in $\boldsymbol{a}_d$ that were included were due to gravitational effects, caused by mars deformation relative to a perfect sphere. They are the main perturbations at that distance from

Mars. [11] The perturbations due to Mars' asteroids could also have a significant effect, but as there is yet to be a good model widely referred to in literature they were not included in this preliminary stage. [12] The state-space function was created and solved with *ode45* with a relative and absolute tolerance of 1e-8.

$$\frac{d^2\boldsymbol{r}}{dt^2} + \mu\frac{\boldsymbol{r}}{r^3} = \boldsymbol{a}_d \tag{10}$$

As can be seen on figure 6, the two propagators give equal results, with a difference of 0.044% from the sphere of influence to the point of intersection, which is negligible and most likely caused by computer precision. This is to be expected, as due to the high energy of the hyperbolic orbit, the magnitude of velocities are very large and therefore the effect of perturbations is relatively smaller than in other orbits of similar altitude but less energy, such as LEO. In addition, as capture by the Martian gravitational force happens in a short window of time, oscillations due to perturbations with a high period would not be observed either. As this shows that the unperturbed model is accurate to a high degree, the previously calculated value of Δ can be used to define the interplanetary trajectory.

4 Integration and Interfacing

4.1 Patching to Interplanetary Trajectory

After the hyperbolic orbit was defined as a function of FPA, the next step was to patch it with the interplanetary trajectory, which will define the remainder of the orbital elements: argument of perigee (ω), right ascension of the ascending node (Ω) and inclination (i). It is interesting to note that Δ is equivalent to the magnitude of the B-plane vector, which is the vector defining the plane orthogonal to the hyperbolic trajectory plane and the initial excess velocity vector. This gives the most flexibility as the B-plane can be rotated to fit the optimal trajectory, whilst ensuring the target FPA is achieved. Further details on the patching procedure may be found in [6]. The final orbital elements can be seen in table 5, as well as the time of arrival.

Table 5: Remainder of orbital elements and time of arrival

Orbital element	Value
i (deg)	75.00
Ω (deg)	155.10
ω (deg)	349.43
Time of arrival	09/10/2031 8:51:46

4.2 Input to Aerothermal team's model

Now that the hyperbolic trajectory is fully defined by the orbital elements, the elements must be converted into the required inputs by the aerothermal team. This includes 3D vectors of radius and velocity, as well as latitude, longitude and heading angle.

It is important to note that the orbital elements are defined with respect to the Mars-Centered Mars Mean Equator and IAU-Vector of Date frame, at the J2000, as shown by figure 5. This is a widely used, non-inertial, coordinate system. Whilst our mission will happen after 2025, and therefore it is expected for us to use J2050, some of the tools used by the Interplanetary team are yet to be updated to J2050. As it is very close to 2025 it was deemed that the error in longitude would be very small, for which an error would have a small effect on the Aerothermal model. To change from the perifocal frame, the vectors were transformed by multiplication with A, shown in equation 11

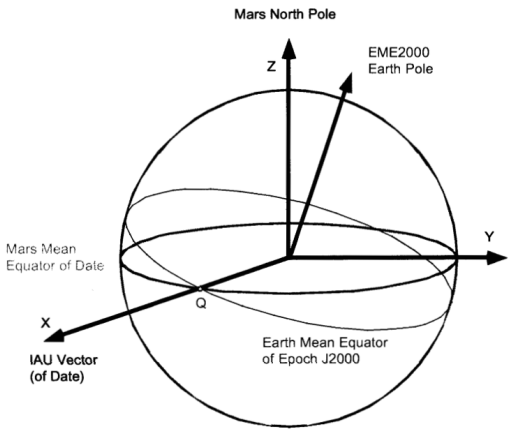


Figure 5: Graphical representation of Centered Mars Mean Equator and IAU-Vector of Date frame. [3]

$$A = \begin{bmatrix} cos(\Omega)cos(\omega) - sin(\Omega)sin(\omega)cos(i) & -cos(\Omega)sin(\omega) - sin(\Omega)cos(\omega)cos(i) & sin(\Omega)sin(i) \\ sin(\Omega)cos(\omega) + cos(\Omega)sin(\omega)cos(i) & -sin(\Omega)sin(\omega) + cos(\Omega)cos(\omega)cos(i) & -cos(\Omega)sin(i) \\ sin(\omega)sin(i) & cos(\omega)sin(i) & cosi \end{bmatrix} \tag{11}$$

Both the perturbed and unperturbed trajectory are plotted in figure 6. In order to calculate latitude and longitude, the units must first be changed so that the x axis points in the direction of the prime meridian of Mars. This can be done by transforming from the Mars-Centered Mars Mean Equator and IAU-Vector of Date frame into the Mars-Centered Mars Mean Equator and Prime Meridian Date frame. This can be done using equation 12 which gives you the angle by which the frame must be rotated, where d is time in days since epoch J2000 [3].

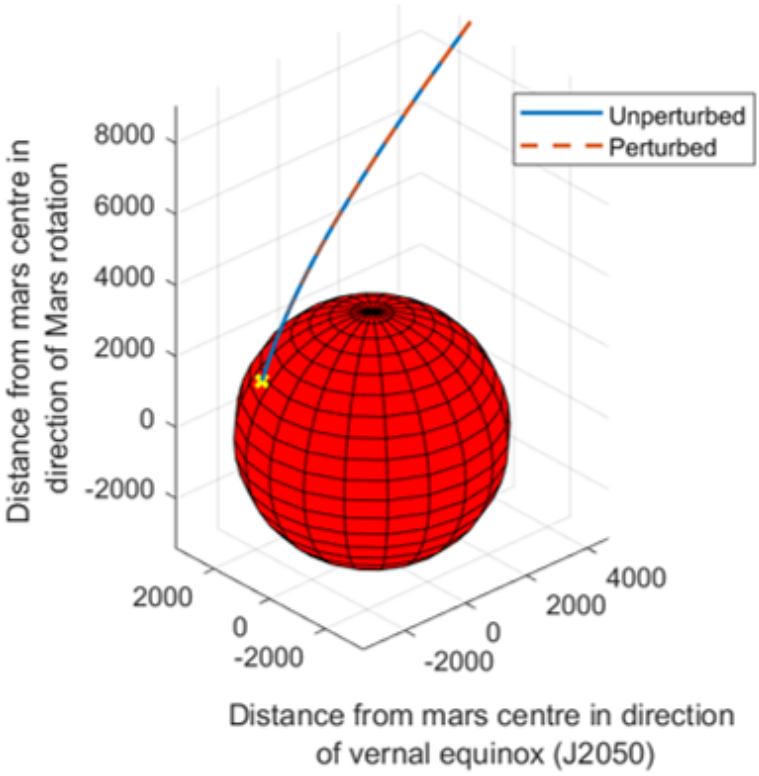


Figure 6: Hyperbolic insertion showing perturbed and unperturbed trajectory

$$W = 176.901 + 350.8919830d \tag{12}$$

Once the frame has been rotated, latitude δ and longitude λ can be calculated using equations 13 and 14. Longitude is affected by quadrant ambiguity, and therefore must be subtracted from 2π if r_y is negative. Once δ has been calculated the velocity azimuth A , also known as the heading angle, can be calculated by equation 15. [13]

$$sin(\delta) = \frac{r_z}{r} \tag{13}$$

$$cos(\lambda) = \frac{r_x}{rcos(\delta)} \tag{14}$$

$$sin(A) = \frac{cos(i)}{cos(\delta)} \tag{15}$$

The calculated values may be found in table 6.

Table 6: Insertion coordinates for aerothermal model.

Angle (°)	Value
δ	34.88
λ	33.06
A	160.16

4.3 Post Aerocapture: Transforming output to orbital elements

After the aerocapture model, the radius vector and velocity vector outputs were transformed into orbital elements to make sure that all of the requirements are met. The output is in the same frame as the input, so the orbital elements can be calculated from the vectors in the usual way, except that W in equation 12 was substracted from the right ascension of the ascending node, to place in a non-rotating frame. The orbital elements of the final orbit can be found in table 7. It can be observed that $i > 70$ meeting the final orbit requirements.

Table 7: Orbital elements after aerocapture

Orbital element	Value
a	2.8095E+03
e	0.7761
i	75.74
Ω	88.57
ω	312.84
θ	208.49

Finally, the burn required for the circularisation of the orbit at apogee can be calculated by comparing the velocities of the circular orbit to the elliptical orbit outputted at its apogee ($1000km$). The value for the burn may be found in table 8, compared to the sum of the equivalent burns that would have been required without the use of aerocapture. As can be observed, the current fuel savings are not as high as expected. This is because due to the model mismatch described in section 5.2, the apoapsis is 600km higher than desired, which leads to the need of a burn in the radial direction. Ideally, the objective is to optimise the aerothermal model so that the eccentricity is minimised, which happens when the semi-major axis is the average between the height of the atmosphere and 1000km. This value is expected to be reached with further iterations.

Table 8: Table showing the total delta-V needed for our current iteration, a potential better optimization, and no aerocapture

Element	Aerocapture	Burn at hyperbola	Refined aerocapture
h_a (km)	1600.4530	N/A	1000
ΔV_p (km/s)	1.5477	-1.6657	0.1780
ΔV_r (km/s)	-1.3366	2.9356	0.0000
ΔV_{tot} (km/s)	2.0450	3.3752	0.1780
γ (°)	-40.8131	150.4288	0

5 Discussion

As outlined previously in sections 3 and , the conics were patched in conjunction with the aerothermal model to achieve a trajectory that maximises accuracy of atmospheric insertion whilst minimizing the fuel required. This section will discuss its shortcomings and next steps.

5.1 Uncertainty propagation throughout the trajectory and its potential effect on mass estimates

Whilst our model suggests that using aerocapture can lead to significant mass savings in terms of fuel and lead to an increase in payload mass, the reality is that this is dependent on the uncertainty found within the model. This can be illlustrated using the final circularisation burn. If the uncertainty is high, it could be that the apogee does not match the expected altitude. If this value is higher, a burn which is not solely perpendicular, and therefore much more expensive fuel-wise, would need to be used. Similarly, if the apogee is lower the semi-major axis will need to be raised with a burn, which is more expensive than the periapsis raise. Therefore the higher the uncertainty, the more fuel that must be carried in case it is required for correction maneuvers. If the uncertainty is too high, the weight of fuel for correction maneuvers could approach the weight of fuel that would be needed for the traditional method. Even if this fuel does not burn, it decreases the mass fraction of the payload, and could therefore make the mission less successful.

Whilst it could be argued that the orbital model's uncertainty is decoupled from the aerocapture model, and will therefore lead to the same fuel for correction maneuvers regardless of whether the novel technology is used or not, in reality this is highly dependent on the sensitivity of the aerothermal model to velocity and FPA inaccuracies. Ideally, this could be studied by a Monte-Carlo simulation studying the apoapsis of the post aerocapture orbit and introducing disturbances at different stages. Whilst this was initially considered, the combination of the aerocapture and orbital code takes about 30 minutes to run, so it wouldn't have finished in the time frame. In addition, the atmosphere was calculated using MarsGRAM, and if the latitude varies by some degrees it will need to be recalculated, meaning there would be a bottleneck in the simulation.

For the next steps of this feasibility study, it would be imperative to find a good method to quantify the uncertainty in the system. It is important to understand that until there is no quantitative measure for uncertainty computed for the whole model, the results must be taken with a pinch of salt.

5.2 Inter-dependency of variables and consequent mismatch between models

As stated in section 2.2, the approach taken considered the dependency of variables to be linear. In reality, there is many interdependencies between variables and therefore functions, as shown by figure 7.

One of the main inter-dependencies is that the heading angle, latitude and longitude, change the dynamics of the aerothermal model, which changes the value for the optimal FPA. This will lead to a mismatch between the new optimal FPA and the one the hyperbolic trajectory is targetting. Additionally, the Δ which the interplanetary trajectory needs to meet will change the excess velocity and the time of arrival. The time of arrival could change the atmospheric conditions [14], just like

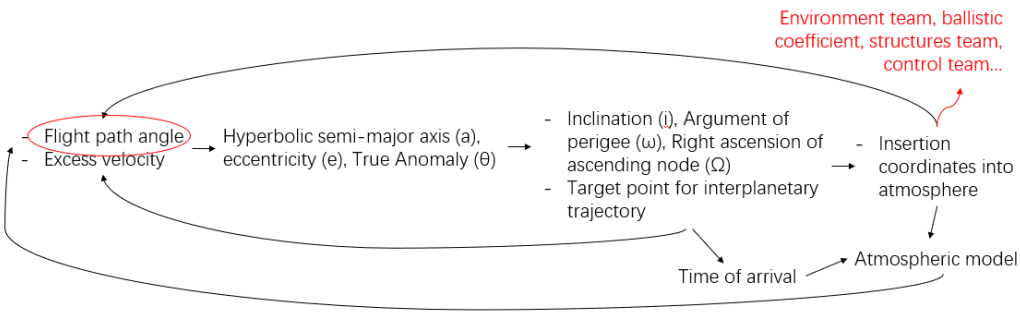


Figure 7: Diagram showing initial approach (straight lines) compared to current identified interdependencies;.

the insertion coordinates, which again effects the optimal FPA that should targetted. Any changes in these variables could have ramifications in the design of all the teams.

A solution that was devised for this was to use matching FPAs to achieve a converged solution. The idea is that even though all the variables are interlinked, as FPA has been used as input it can also be used to see when all of the values have stopped changing. Whilst the changes in atmosphere cannot be looped in the code, they are expected to be very small if the latitude does not change massively, and the change in time is kept within a month of the intial esimate [14]. Therefore a numerical scheme could be used to iterate until a fully optimised solution is reached.

In the end, this was not done for two reasons. The first was simply practical: the iteration takes up to 30 minutes, so depending on the numerical method used convergence could take days. Secondly, it was not considered critical, as currently there is a mismatch of flight path angle of 0.4 degrees, well within the stability corridor discussed in detail in [15]. This, however, does not account for potential insertion inaccuracy, so the iteration should be completed in the next steps for a more robust design.

References

[1] Howard D. Curtis. *Orbital Mechanics for Engineering Students: Revised Reprint*. Butterworth-Heinemann, Aug 31, 2020. ISBN 9780128240250. URL https://www.vlebooks.com/vleweb/product/openreader?id=none&isbn=9780323853453.

[2] Rob Hewson and Davide Amato. Orbital mechanics lecture notes. 2022. URL https://bb.imperial.ac.uk/webapps/blackboard/execute/content/file?cmd=view&content_id=_2413575_1&course_id=_31858_1.

[3] R. Mase and R. Roncoli. Update to mars coordinate frame definitions. Technical report, Jan 1, 1999. URL http://ntrs.nasa.gov/search.jsp?R=20060034233.

[4] Mohit Tank. Mission architecture and future roadmap for aegis . June 20, 2022.

[5] Aaditi Vaze. Mission architecture and interplanetary mission design analysis . June 20, 2022.

[6] Luca Mazzotta. Design and optimization of the interplanetary transfer orbit . Jun 20, 2022.

[7] ESA ESTEC. Aerocapture demonstration mission - expro plus, Dec 1, 2021.

[8] James R. Wertz. *Space mission engineering: The New SMAD*. Microcosm Press, Hawthorne, CA, 2011. ISBN 9781881883159.

[9] John E. Prussing and Bruce A. Conway. *Orbital mechanics*. Oxford Univ. Press, New York [u.a.], 1. print. edition, 1993. ISBN 0195078349.

[10] Richard H. Battin. *An introduction to the mathematics and methods of astrodynamics*. American Inst. of Aeronautics and Astronautics, Reston, VA, rev. ed., [nachdr.] edition, 1999. ISBN 1563473429.

[11] V. U. J. Nwankwo and S. K. Chakrabarti. Analysis of planetary and solar-induced perturbations on trans-martian trajectory of mars missions before and after mars orbit insertion. *Indian Journal of Physics*, 89(12):1235–1245, May 28, 2015. doi: 10.1007/s12648-015-0705-9. URL https://link.springer.com/article/10.1007/s12648-015-0705-9.

[12] S. Mouret, J. L. Simon, F. Mignard, and D. Hestroffer. The list of asteroids perturbing the mars orbit to be seen during future space missions. *Astronomy and astrophysics (Berlin)*, 508(1):479–489, Dec 2009. doi: 10.1051/0004-6361/200810979. URL https://api.istex.fr/ark:/67375/80W-8NNKWW9B-V/fulltext.pdf.

[13] Nicola Bellomo and Ashish Tewari. *Atmospheric and Space Flight Dynamics*. Birkhäuser Boston, Boston, 2007. ISBN 9780817643737. URL https://ebookcentral.proquest.com/lib/[SITE_ID]/detail.action?docID=3062930.

[14] Erikas Kymantas. Flight-control architecture analysis atmospheric driven trajectory study with mars-gram 2010 . June 20, 2022.

[15] Segundo Urraza Atue. Analysis of atmospheric and space flight dynamics and aerocapture trajectory optimization . June 20, 2022.